

Proof. Let x and y be vectors in an inner-product space $(V, \langle \cdot, \cdot \rangle)$. The norm induced by the inner product is

$$\|v\| = \sqrt{\langle v, v \rangle} \quad (v \in V). \quad (1)$$

1. Reduction to the squared inequality. Both $\|x + y\|$ and $\|x\| + \|y\|$ are non-negative, hence the triangle inequality

$$\|x + y\| \leq \|x\| + \|y\| \quad (2)$$

is equivalent to

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (3)$$

2. Expansion of $\|x + y\|^2$. Using bilinearity of the inner product and $\langle y, x \rangle = \overline{\langle x, y \rangle}$,

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \end{aligned} \quad (4)$$

3. Expansion of the right-hand side of (3).

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2. \quad (5)$$

4. Application of Cauchy-Schwarz. The Cauchy-Schwarz inequality gives

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (6)$$

Since $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$, we obtain

$$2 \operatorname{Re} \langle x, y \rangle \leq 2\|x\| \|y\|. \quad (7)$$

5. Comparison of (4) and (5). Insert the estimate (7) into (4):

$$\begin{aligned} \|x + y\|^2 &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2. \end{aligned}$$

Thus (3) holds.

6. Return to the original inequality. Both sides of (3) are non-negative; taking square roots yields (2):

$$\boxed{\|x + y\| \leq \|x\| + \|y\|}.$$

7. Equality condition. Equality in the chain above occurs precisely when equality holds in both $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$ and in the Cauchy-Schwarz inequality (6).

- $\operatorname{Re}\langle x, y \rangle = |\langle x, y \rangle|$ means that $\langle x, y \rangle$ is a non-negative real number.
- Equality in Cauchy–Schwarz holds iff x and y are linearly dependent, i.e. $y = \lambda x$ for some scalar $\lambda \in \mathbb{C}$.

Combining the two requirements forces λ to be a *real non-negative* scalar. Hence, for non-zero vectors,

$$\|x + y\| = \|x\| + \|y\| \iff y = \lambda x \text{ with } \lambda \geq 0.$$

The statement also holds trivially when $x = 0$ or $y = 0$.

Consequently, the triangle inequality is proved, and equality occurs exactly when the vectors are collinear and point in the same (or zero) direction. $\square$ $\square$